



Improving the Shutdown Switch Positioning on 793D Mining Trucks

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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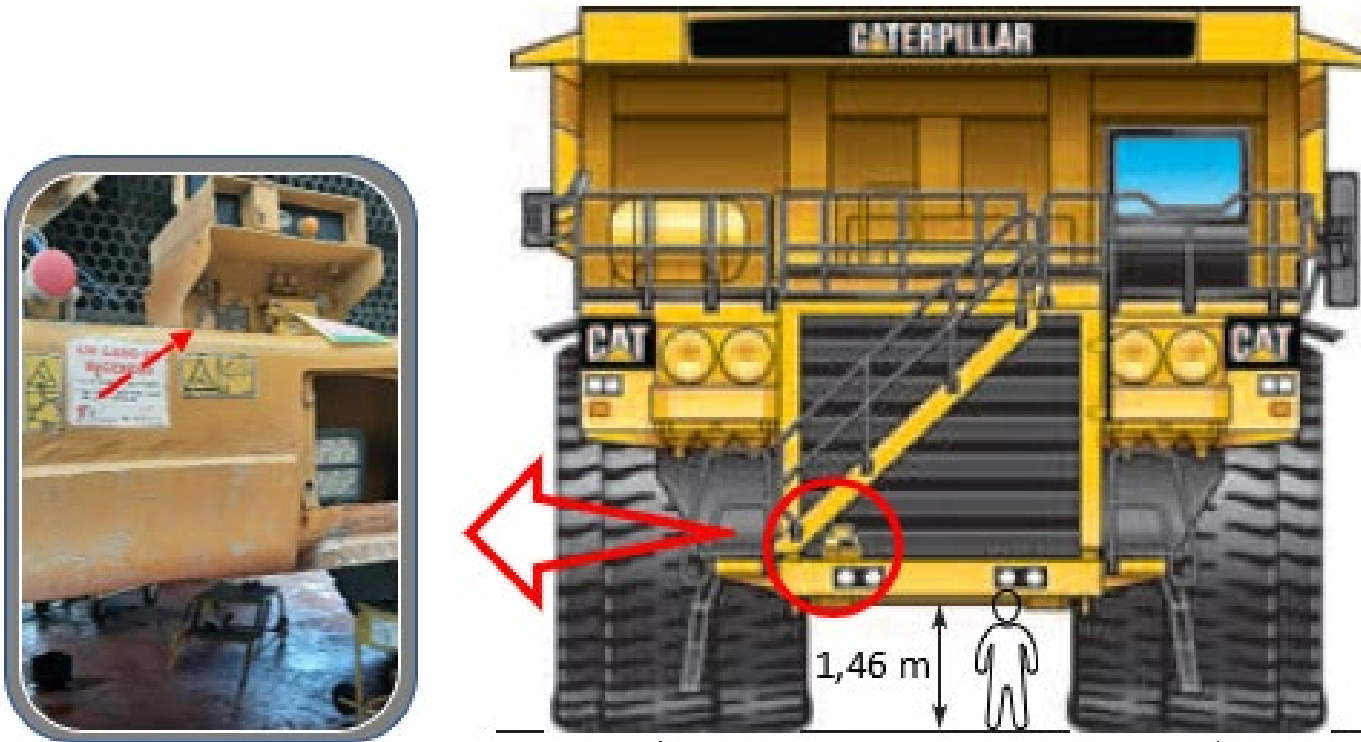
December 2022
1.01-221219-1

1.0 Introduction

This Best Practice reduces possible work and material accidents by changing the position of this switch reducing the height by approximately 30 cm (11,8" Inches) in relation to original position.

In the Brazil North region, the median height of operators is frequently under the world average, sometimes the support structure is required for improving the ergonomics condition and mitigating risks for the employees.

Implementing this Best Practice makes it possible for more people to access the shutdown switch safely and more efficiently.

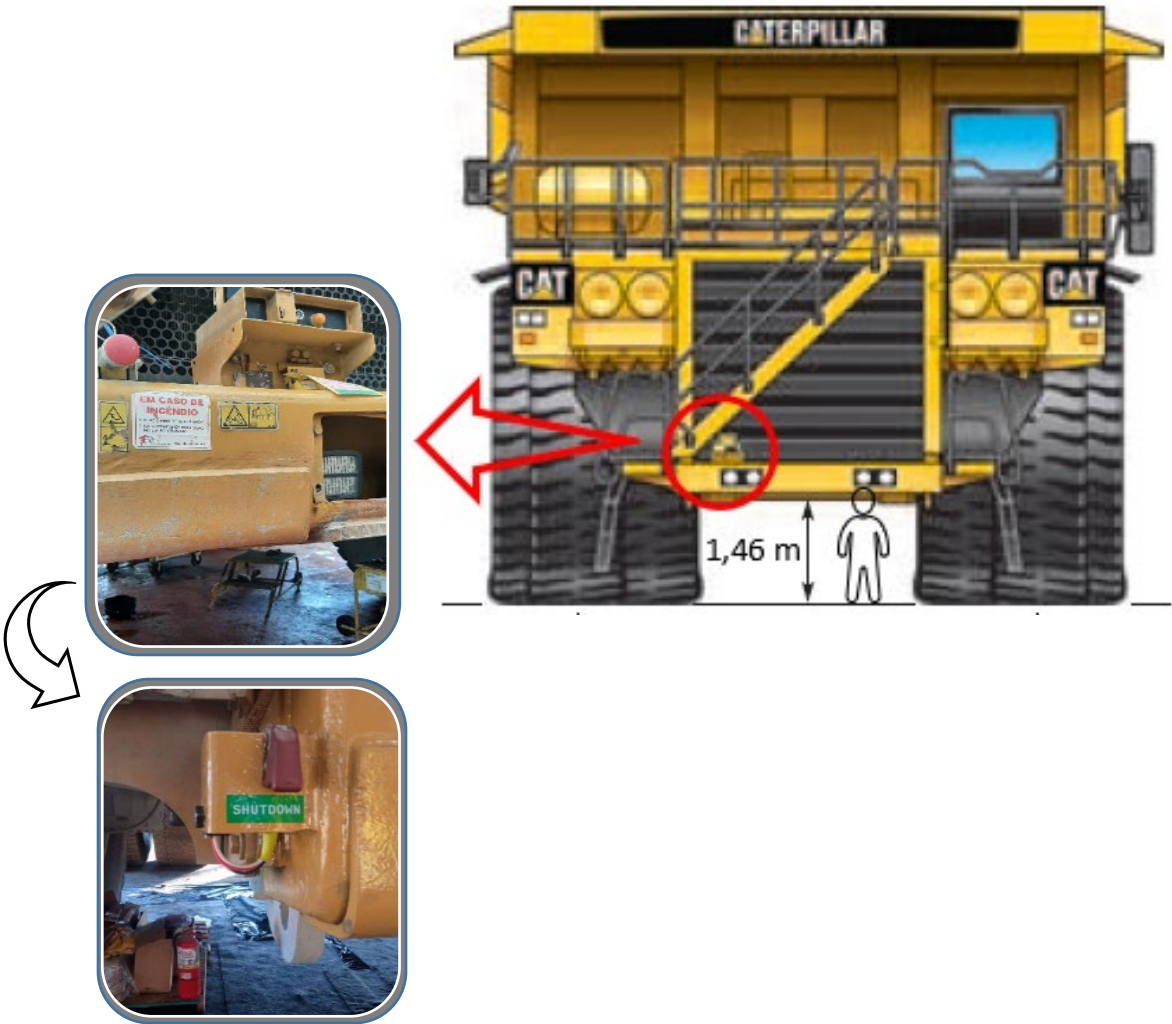


Original position of the Shutdown Switch – Source: SOTREQ/Giesi Nascimento.

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2.0 Best Practice Description

For reducing the difficulty of accessing the switch by the operator, SOTREQ moved the shutdown switch to the front right side (near the access ladder) of the CAT 793D mining truck, thus eliminating the use of any auxiliary device to access the switch.

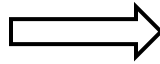


New shutdown switch position – Source: SOTREQ/Giesi Nascimento.

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3.0 Implementation Steps

The implementation of this Best Practice is described below. Contact **SOTREQ** to request additional information about specific details.



After removal, it will be necessary a Harness to connect the switch to the Electrical Circuit. Approximately 1 m of 3 mm thick 03 threads wire will be required.



To connect the switch to the assembled cable, use **02 (03 Threads) Deutsch** connectors.

On the front of the equipment, weld the switch protection box to the chassis spar and connect the harness to the switch.

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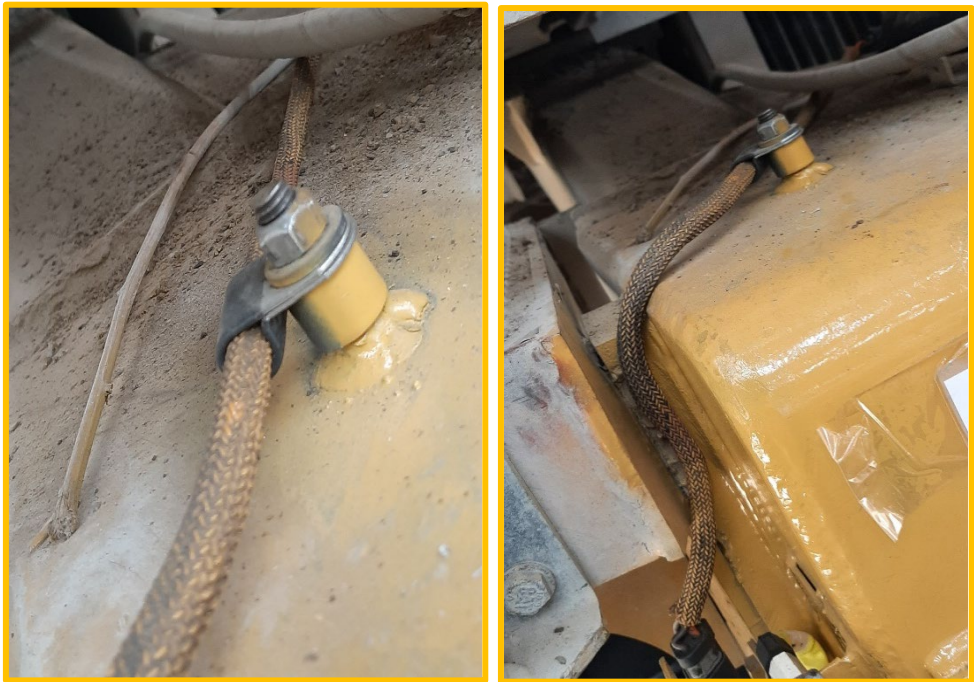


Before/After Implementation – Source: SOTREQ/Giesi Nascimento.



Fix the harness to the support with 01 clamp (PN 130-5300), 01 bolt (PN 8T-4136), 01 Washer (PN 8T-4121), 01 M10 nuts.

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Harness Fixing Process Before/After - Source: **SOTREQ S/A**

Make the connection of the 03 Threads connector with the equipment's electrical circuit.

Test activating the switch to verify the functionality.



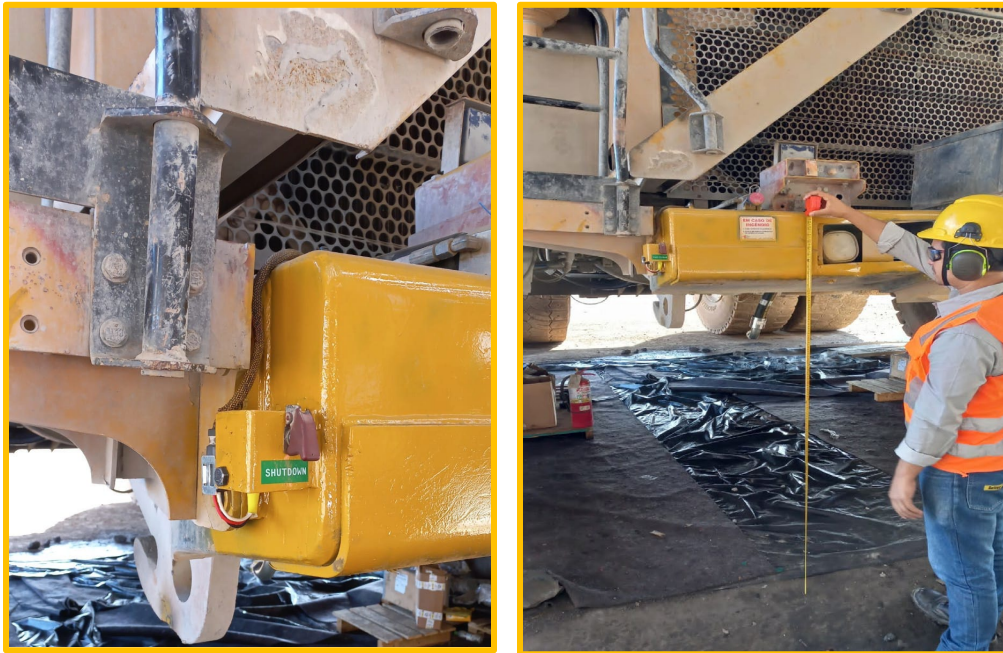
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4.0 Benefits

This change position, reduce the distance range by 16% (1.88 m to 1.57 m) by the operator/technician who is working on the maintenance of the equipment.

Additionally, a time reduction (New Time: Approximately 30 seconds) was observed after implementation of this Best Practice:

- **Productivity** – Time Reduction for Switch Actuation.
- **Personal Safety** – Engine Shutdown process simplified in risk situations.

Testing after installation:

- Operator Height 1.51 m.

Before the change: Difficulty accessing and viewing the switch

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After the change: easy visualization and no difficulty in activating the switch.



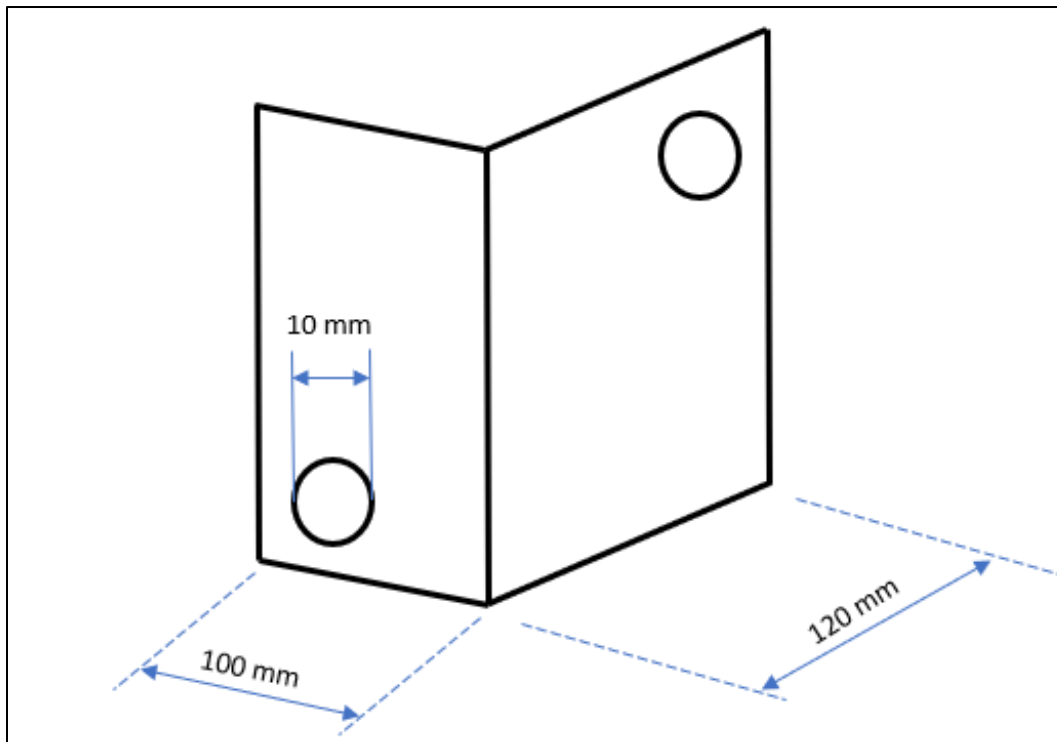
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5.0 Resources Required

Qty.	Unity	Descrição
1	Unity	Fixing Toggle
1	Unity	5P-9299 Clip
2	Unity.	8T-4136 Bolt
2	Unity.	M10 Sow
2	Unity.	8T-4121 Hard Washer
1	Unity.	Male Connector (3 Threads)
1	Unity.	Female Connector (3 Threads)
1	Unity.	3E-5169 Switch Guard
1	Unity.	158-6634 Switch
1	Unity.	130-5300 Clip
1	M	3,0 mm Wire (3 Threads)

6.0 Supporting Attachments / References

For additional Information about mounting please contact SOTREQ Engineering Team.



3D Drawings: Fonte – SOTREQ/Giesi Nascimento.

7.0 Related Best Practices

N/A

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8.0 Acknowledgements

The Sotreq Team

Special Thanks To: **Doug Friedrich** (Caterpillar Inc.)

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